Philosophy of AI

Course Description

Artificial Intelligence is one of the most exciting and fastest-moving technological fields today. This course examines philosophical questions related to AI. We will explore big-picture questions such as whether AIs can be conscious, whether we can control them, whether there might be an intelligence explosion. What, if anything, distinguishes intelligence from computation, prediction, or pattern recognition? How should we think about AI agency, autonomy, and responsibility? We will also consider ethical and social issues involving large language models, transparency and explainability, the ethical status of robots, and the ethics of deploying AI in high-stakes domains. No programming experience required.

Learning Objectives

The goals of this course are to (i) develop a basic understanding of the key concepts in AI, (ii) reflect on assumptions about what AI is and what it means for humans, (iii) develop a basic understanding of major philosophical theories of mind and apply them to the AI context, (iv) develop and improve skills in analyzing, formulating, explaining, and critically evaluating arguments and in how to process and learn from critical feedbacks, (iv) learn how to read and think critically and write clearly.

Outline

1. Introduction to the Course

Tim Urban, "The AI Revolution", Section: The Road to Superintelligence.

Buchanan, "A (very) brief history of artificial intelligence" in *AI Magazine,* 26 (2005); p. 53-56.

2. Symbolic v.s. Subsymbolic AI

Mitchell, *Artificial Intelligence*, Chapter 1.

Supplementary: Allen Newell and Herbert Simon, *Computer Science as Empirical Inquiry: Symbols and Search* (1975).

Supplementary: John Haugeland, *Semantic Engines* (Introduction to *Mind Design*, First Edition).

3. Machine Learning and Neural Networks

Mitchell, *Artificial Intelligence*, Chapter 2.

Video (slightly more mathematical): 3Blue1Brown, But what is a neural network?

Optional: play with https://playground.tensorflow.org/

Supplementary: 3Blue1Brown, Gradient descent, how neural networks learn.

Supplementary: 3Blue1Brown, Backpropagation, intuitively.

4. Large Language Models (LLMs) and Transformers

Video: 3Blue1Brown, Transformers, the tech behind LLMs.

Reading Alternative: Ben Levinstein, A Conceptual Guide to Transformers: Part I, Architecture of the Transformer Model. Supplemented with a video: 3Blue1Brown, Large Language Models explained briefly.

Supplementary: Mitchell, *Artificial Intelligence*, Chapter 11-13.

Supplementary: Buckner, "Deep learning: A philosophical introduction" (2019).

5. Transparency and Interpretability

Creel, K. A. (2020). "Transparency in complex computational systems" in *Philosophy of Science*, 87(4), 568- 589.

Zednik (2019). "Solving the Black Box Problem".

Supplementary: Lipton (2019) "The Mythos of Model Interpretability".

Supplementary: Nguyen, "Transparency is Surveillance".

6. The Chinese Room Argument and the Turing Test

John Searle, "Mind, Brains, and Programs" in *The Behavior and Brain Sciences* (1980) 3, 417-457.

Alan Turing, "Computing Machinery and Intelligence", in *Mind* (1950), p. 433-460.

Supplementary: Daniel Dennett, "Can Machines Think?"

Supplementary: Mitchell, *Artificial Intelligence*, Chapter 3.

Supplementary: Susan Sterrett, Turing's Two Tests for Intelligence .

7. Consciousness

David Chalmers, "Could a Large Language Model be Conscious?" (2023). arXiv preprint arXiv:2303.07103

Eric Schwitzgebel, "Global Workspace Theories and Higher Order Theories" in *AI and Consciousness: A Skeptical Overview*, forthcoming in the *Cambridge Elements* series; Chapter 8, Intro & Section 1-2 (skip paragraphs 3-5 in Section 1).

Supplementary: McDermott, D. (2007). "Artificial intelligence and consciousness" in *the Cambridge handbook of consciousness*, 117-150.

Supplementary: Tim Maudlin, "Computation and Consciousness" in the Journal of Philosophy, Vol. 86, No. 8 (Aug., 1989), p. 407-432.

8. Emergence in AI

David C. Krakauer, Melanie Mitchell, John W. Krakauer, "Large language models and emergence: a complex systems perspective" in Phil. Trans. R. Soc. A 14 May 2026; 384 (2320): 20250014.

Supplementary: Mark Bedau, "Weak Emergence" (1997).

9. Superintelligence and Singularity

David Chalmers, "The Singularity: A Philosophical Analysis" in the *Journal of Consciousness Studies* 17(9-10) 2010, p. 7-65; Section 1-8.

Supplementary: Irving John Good, "Speculations Concerning the First Ultraintelligent Machine", in *Advances in Computers*, vol. 6 (Elsevier, 1966), p. 31-88.

Supplementary: Cappelen, Herman, Goldstein, Simon, and Hawthorne, John (2025), "AI Survival Stories: a Taxonomic Analysis of AI Existential Risk" in Philosophy of AI 1 (1):1-18.

10. AI for Science

Bradshaw, Langley, and Simon (1983) "Studying Scientific Discovery by Computer Simulation"

Korb (2004) Machine Learning as Philosophy of Science

Supplementary: Collins 1989 "Computers and the Sociology of Scientific Knowledge"

Supplementary: Iten et al. 2020 "Discovering physical concepts with neural networks"

11. Alignment

Gabriel, I. (2020), "Artificial Intelligence, Values, and Alignment" in *Minds and Machines*, 30 (3), 411-437.

Supplementary: Russell, S. (2019). *Human compatible: Artificial intelligence and the problem of control*. Penguin.

Supplementary: Wallach, Wendell, and Shannon Vallor, "Moral Machines: From Value Alignment to Embodied Virtue", in S. Matthew Liao (ed.), *Ethics of Artificial Intelligence* (New York, 2020).

Supplementary: Zhi-Xuan, T., Carroll, M., Franklin, M. et al. "Beyond Preferences in AI Alignment" in *Philosophical Studies* 182, 1813–1863 (2025).

Supplementary video: Writing AI Constitutions (Joe Carlsmith, Anthropic).

Supplementary: Carlsmith, Joe. "Scheming AIs: Will AIs fake alignment during training in order to get power?" arXiv preprint arXiv:2311.08379 (2023).

12. Algorithmic Fairness and Bias in AI

Fazelpour & Danks (2021) "Algorithmic Bias: Senses, Sources, Solutions"

Gabbrielle Johnson, "Algorithmic Bias: On the Implicit Biases of Social Technology." Synthese (June 20, 2020, online first)

Supplementary: Grant (2023) "Equalized Odds is a Requirement of Algorithmic Fairness"

Supplementary: Creel & Hellman (2021) The Algorithmic Leviathan: Arbitrariness, Fairness, and Opportunity in Algorithmic Decision-Making Systems

13. Moral Machines

Railton, Peter, "Ethical Learning, Natural and Artificial" in S. Matthew Liao (ed.), *Ethics of Artificial Intelligence* (New York, 2020).

Supplementary: Kamm, F. M., "The Use and Abuse of the Trolley Problem: Self-Driving Cars, Medical Treatments, and the Distribution of Harm" in S. Matthew Liao (ed.), Ethics of Artificial Intelligence (New York, 2020).

14. Privacy in the age of Big Data

Matzner (2013). Why Privacy is Not Enough Privacy in the Context of "Ubiquitous Computing" and "Big Data".

Supplementary: Nissenbaum, "Privacy as Contextual Integrity".

15. Healthcare

Bjerring, J. C., & Busch, J. (2021). "Artificial intelligence and patient-centered decision-making" in *Philosophy & Technology*, 34, 349-371.

Supplementary: Schönberger, D. (2019). Artificial intelligence in healthcare: a critical analysis of the legal and ethical implications. International Journal of Law and Information Technology, 27(2), 171-203.

Supplementary: Kasirzadeh Algorithmic Fairness and Structural Injustice: Insights from Feminist Political Philosophy